

$$\textcircled{1} \begin{pmatrix} 1 & 4 & -1 & | & 6 \\ 2 & 5 & -7 & | & -9 \\ 3 & -2 & 1 & | & 2 \end{pmatrix} \quad R_2 = 2R_1 + R_2 \quad \begin{pmatrix} 1 & 4 & -1 & | & 6 \\ 0 & -3 & -5 & | & -2 \\ 3 & -2 & 1 & | & 2 \end{pmatrix}$$

$$R_3 = -3R_1 + R_3 = \begin{pmatrix} 1 & 4 & -1 & | & 6 \\ 0 & -3 & -5 & | & -2 \\ 0 & -14 & 4 & | & -6 \end{pmatrix}$$

$$R_3 = -3R_3 + 14R_2 = \begin{pmatrix} 1 & 4 & -1 & | & 6 \\ 0 & -3 & -5 & | & -2 \\ 0 & 0 & -82 & | & 20 \end{pmatrix}$$

$$82z = 20$$

$$z = 10/41$$

$$-3y - 5(10/41) = -2$$

$$2/41 + 50/41 = 3y$$

$$132/123 = y$$

$$44/41 = y$$

$$\textcircled{2} \begin{pmatrix} 2 & 3 & 1 & | & 1 \\ 6 & -2 & -1 & | & -14 \\ 3 & 1 & -1 & | & -1 \end{pmatrix} \quad R_3 = 2R_3 \quad \begin{pmatrix} 2 & 3 & 1 & | & 1 \\ 6 & -2 & -1 & | & -14 \\ 6 & 2 & -2 & | & -2 \end{pmatrix}$$

$$R_2 = R_2 - R_3 \quad \begin{pmatrix} 2 & 3 & 1 & | & 1 \\ 0 & -4 & 1 & | & -12 \\ 6 & 2 & -2 & | & -2 \end{pmatrix} \quad R_1 = 3R_1 \quad \begin{pmatrix} 6 & 9 & 3 & | & 3 \\ 0 & -4 & 1 & | & -12 \\ 6 & 2 & -2 & | & -2 \end{pmatrix}$$

$$R_3 = R_3 - R_1 = \begin{pmatrix} 6 & 9 & 3 & | & 3 \\ 0 & -4 & 1 & | & -12 \\ 0 & -7 & -5 & | & -5 \end{pmatrix} \quad \begin{matrix} R_3 = \frac{1}{6}R_3 \\ R_2 = 7R_2 \end{matrix} \quad \begin{pmatrix} 6 & 9 & 3 & | & 3 \\ 0 & -28 & 7 & | & -84 \\ 0 & 28 & 20 & | & 20 \end{pmatrix}$$

$$R_3 = R_3 + R_2 = \begin{pmatrix} 6 & 9 & 3 & | & 3 \\ 0 & -28 & 7 & | & -84 \\ 0 & 0 & 27 & | & -64 \end{pmatrix} = R_2 = R_2 / 7 \quad \begin{pmatrix} 6 & 9 & 3 & | & 3 \\ 0 & -4 & 1 & | & -12 \\ 0 & 0 & 27 & | & -64 \end{pmatrix}$$

$$27z = -64 \Rightarrow z = \boxed{-64/27}$$

$$-4y + (-64/27) = -12$$

$$-4y = -12 + (64/27) \Rightarrow -4y = -388/27 \Rightarrow y = -388/27 / -4/1$$

$$= \boxed{97/27}$$

$$= -388/$$

$$6x + 9(97/27) + 3(-64/27) = 3$$

$$2x + 3(97/27) + (-64/27) = 1$$

$$2x + 3(97/27) - 64/27 = 1$$

$$2x + (97/9) - (64/27) = 1$$

$$2x = 1 - (97/9) + (64/27)$$

$$2x = -200/27$$

$$x = -200/27 / 2/1$$

$$x = \boxed{-100/27}$$

Jordan

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & -4 & 1 & -12 \\ 0 & 0 & 27 & -64 \end{array} \right] \quad f_2 = -27f_1$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & 108 & -27 & 324 \\ 0 & 0 & 27 & -64 \end{array} \right] \quad f_2 = f_2 + f_3$$

$$= \left[\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 0 & 108 & 0 & 260 \\ 0 & 0 & 27 & -64 \end{array} \right] \quad \begin{array}{l} f_1 = 27f_1 \\ f_2 = f_2/2 \end{array}$$

$$\left[\begin{array}{ccc|c} -54 & -81 & -27 & -27 \\ 0 & 108 & 0 & 260 \\ 0 & 0 & 27 & 270 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} -54 & -81 & -27 & -27 \\ 0 & 108 & 0 & 260 \\ 0 & 0 & 27 & 270 \end{array} \right] \quad R_1 = R_1 + R_3$$

$$\left[\begin{array}{ccc|c} -54 & -81 & 0 & 243 \\ 0 & 108 & 0 & 260 \\ 0 & 0 & 27 & 270 \end{array} \right]$$

$$\begin{aligned} R_1 &= R_1 \times 108 \\ R_2 &= -81 R_2 \end{aligned} \quad = \left[\begin{array}{ccc|c} 5832 & 8748 & 0 & \\ 0 & -8748 & 0 & \\ 0 & 0 & 27 & 270 \end{array} \right]$$

$$R_1 = R_1 + R_2 \quad \left[\begin{array}{ccc|c} 5832 & 0 & 0 & 1584 \\ 0 & -8748 & 0 & -21060 \\ 0 & 0 & 27 & 270 \end{array} \right]$$

$$\begin{aligned} R_1 &= R_1 / 27 \\ R_2 &= R_2 / 81 \end{aligned} \quad \left[\begin{array}{ccc|c} 216 & 0 & 0 & 58 \\ 0 & 108 & 0 & 260 \\ 0 & 0 & 27 & 270 \end{array} \right]$$

$$\text{sd } \{22, 260, 270\}$$

$$\textcircled{3} \begin{bmatrix} 2 & 3 & 1 & | & 3 \\ 10 & -8 & -9 & | & 0 \\ 4 & 4 & -3 & | & 2 \end{bmatrix} \xrightarrow{SF1} \begin{bmatrix} 10 & 15 & 5 & | & 15 \\ 10 & -8 & -9 & | & 0 \\ 4 & 4 & -3 & | & 2 \end{bmatrix}$$

$$F_2 = F_2 - F_1 \begin{bmatrix} 10 & 15 & 5 & | & 15 \\ 0 & -7 & -14 & | & -15 \\ 4 & 4 & -3 & | & 2 \end{bmatrix} \xrightarrow{F_1/5} \begin{bmatrix} 2 & 3 & 1 & | & 3 \\ 0 & -7 & -14 & | & -15 \\ 4 & 4 & -3 & | & 2 \end{bmatrix}$$

$$2F_1 = \begin{bmatrix} 4 & 6 & 2 & | & 6 \\ 0 & -7 & -14 & | & -15 \\ 4 & 4 & -3 & | & 2 \end{bmatrix} = F_3 = F_3 - F_1 \begin{bmatrix} 4 & 6 & 2 & | & 6 \\ 0 & -7 & -14 & | & -15 \\ 0 & -2 & -5 & | & -4 \end{bmatrix}$$

$$2F_2 = -7F_3 = \begin{bmatrix} 4 & 6 & 2 & | & 6 \\ 0 & -14 & -28 & | & -30 \\ 0 & 14 & 35 & | & 28 \end{bmatrix} \xrightarrow{F_3 = F_3 + F_2} \begin{bmatrix} 4 & 6 & 2 & | & 6 \\ 0 & -14 & -28 & | & -30 \\ 0 & 0 & 7 & | & -2 \end{bmatrix}$$

$$7z = -2 \Rightarrow z = -2/7$$

$$-14y =$$

$$\begin{matrix} F_1/2 \\ F_2/2 \end{matrix} \begin{bmatrix} 2 & 3 & 1 & | & 3 \\ 0 & -7 & -14 & | & -15 \\ 0 & 0 & 7 & | & -2 \end{bmatrix}$$

$$\begin{pmatrix} 2 & 3 & 1 & | & 3 \\ 0 & -7 & -14 & | & -15 \\ 0 & 0 & 7 & | & -2 \end{pmatrix}$$

$$7z = -2 \Rightarrow z = -2/7$$

$$-7y - 14(-2/7) = -15$$

$$-7y - 4 = -15$$

$$-7y = -15 + 4 \Rightarrow -7y = -11$$

$$y = -11/-7$$

$$2x + 3(-11/-7) + (-2/7) = 3$$

$$2x = (31/7) + 3 \Rightarrow 2x = -10/7$$

$$x = -10/7 / 2/1 \Rightarrow x = -5/7$$

Jordan

$$\begin{pmatrix} 2 & 3 & 1 & | & 3 \\ 0 & -7 & -14 & | & -15 \\ 0 & 0 & 7 & | & -2 \end{pmatrix}$$

$$F_3 = 2F_3 \quad \begin{pmatrix} 2 & 3 & 1 & | & 3 \\ 0 & -7 & -14 & | & -15 \\ 0 & 0 & 14 & | & -4 \end{pmatrix} \quad F_2 = F_3 + F_2$$

$$= \begin{bmatrix} 2 & 3 & 1 & | & 3 \\ 0 & -7 & 0 & | & -19 \\ 0 & 0 & 14 & | & -4 \end{bmatrix} \quad F_1 = 14F_3 \quad \begin{bmatrix} 28 & 42 & 14 & | & 42 \\ 0 & -7 & 0 & | & -19 \\ 0 & 0 & 14 & | & -4 \end{bmatrix}$$

$$F_1 = F_1 - F_3 = \begin{bmatrix} 28 & 42 & 0 & | & 42 \\ 0 & -7 & 0 & | & -19 \\ 0 & 0 & 14 & | & -4 \end{bmatrix} \quad F_1/14 \quad \begin{bmatrix} 2 & 3 & 0 & | & 3 \\ 0 & -7 & 0 & | & -19 \\ 0 & 0 & 14 & | & -4 \end{bmatrix}$$

$$\frac{F_1}{3F_2} = \begin{bmatrix} 14 & 21 & 0 & | & 21 \\ 0 & -21 & 0 & | & -57 \\ 0 & 0 & 14 & | & -4 \end{bmatrix} \quad F_1 = F_1 + F_2 \quad \begin{bmatrix} 14 & 0 & 0 & | & -36 \\ 0 & -21 & 0 & | & -57 \\ 0 & 0 & 14 & | & -4 \end{bmatrix}$$

$$\begin{matrix} F_1/2 \\ F_2/-3 \\ F_3/2 \end{matrix} = \begin{bmatrix} 7 & 0 & 0 & | & 18 \\ 0 & 7 & 0 & | & 19 \\ 0 & 0 & 7 & | & -2 \end{bmatrix}$$

$$\text{sol } \{(18, 19, -2)\}$$

④

Gauss

$$\begin{bmatrix} 7 & 3 & -4 & -35 \\ 3 & -2 & 5 & 38 \\ 1 & 1 & -6 & -27 \end{bmatrix}$$

 $f_1 = f_3$

$$\begin{bmatrix} 1 & 1 & -6 & -27 \\ 3 & -2 & 5 & -38 \\ 1 & 1 & -6 & -35 \end{bmatrix}$$

$$f_2 = -3f_1 + f_2 \quad \begin{bmatrix} 1 & 1 & -6 & -27 \\ 0 & -5 & 23 & 119 \\ 7 & 3 & -4 & -35 \end{bmatrix}$$

$$f_3 = -7f_1 + f_3 \quad \begin{bmatrix} 1 & 1 & -6 & -27 \\ 0 & -5 & 23 & 119 \\ 0 & -4 & 38 & 154 \end{bmatrix}$$

$$f_3 = -4f_2 + 5f_3 \quad \begin{bmatrix} 1 & 1 & -6 & -27 \\ 0 & -5 & 23 & 119 \\ 0 & 0 & 98 & 294 \end{bmatrix}$$

$$98z = 294$$

$$z = 294/98 = 3$$

$$-5y + 23(3) = 119$$

$$-5y + 69 = 119$$

$$-5y = 119 - 69$$

$$y = 50/5 = 10$$

$$x = 10 - 6(3) = -27$$

$$x - 10 - 18 = -27$$

$$x - 28 = -27$$

$$x = -27 + 28$$

$$x = 1$$

$$\text{Sol of } (1, 10, 3)$$

Jordan

$$\begin{pmatrix} 1 & 1 & -6 & -27 \\ 0 & -5 & 23 & 119 \\ 0 & 0 & 98 & 299 \end{pmatrix}$$

$$R_2 = -23R_3 + 98R_2 = \begin{pmatrix} 1 & 1 & -6 & -27 \\ 0 & -490 & 0 & 4900 \\ 0 & 0 & 98 & 299 \end{pmatrix}$$

$$-1R_2/490 =$$

$$\begin{pmatrix} 1 & 1 & -6 & -27 \\ 0 & 1 & 0 & -10 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

$$R_1 = R_1 - R_2 = \begin{pmatrix} 1 & 0 & -6 & -17 \\ 0 & 1 & 0 & -10 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

$$R_1 = 6R_3 + R_1 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -10 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

$$\text{sol of } (1, 10, 3)$$

$$\textcircled{5} \begin{pmatrix} 6 & 5 & 2 & 12 \\ 9 & -1 & 4 & 37 \\ 10 & 5 & 3 & 21 \end{pmatrix}$$

$$R_2 = 6R_2 + 9R_1 = \begin{pmatrix} 6 & 3 & 2 & 12 \\ 0 & 33 & -6 & -114 \\ 10 & 5 & 3 & 21 \end{pmatrix}$$

$$R_3 = 10R_1 + 6R_3$$

$$\begin{pmatrix} 6 & 3 & 2 & 12 \\ 0 & 33 & -6 & -114 \\ 0 & 0 & -2 & 6 \end{pmatrix}$$

$$-2z = 6 \Rightarrow z = 6/-2 \Rightarrow z = -3$$

$$33y - 6(-3) = -114$$

$$y = -132/33 \Rightarrow y = -4$$

$$6x + 3y + 2z = 12$$

$$6x + 3(-4) + 2(-3) = 12$$

$$6x - 12 - 6 = 12$$

$$6x - 18 = 12$$

$$6x = 30$$

$$x = 30/6 \Rightarrow x = 5$$

Jordan

$$\left[\begin{array}{ccc|c} 6 & 3 & 2 & 12 \\ 0 & 33 & -6 & -114 \\ 0 & 0 & -2 & 6 \end{array} \right]$$

$$R_1 \leftarrow R_1 + R_3 \quad \left[\begin{array}{ccc|c} 6 & 3 & 0 & 18 \\ 0 & 33 & -6 & -114 \\ 0 & 0 & -2 & 6 \end{array} \right]$$

$$R_2 \leftarrow -3R_3 + R_2 \quad \left[\begin{array}{ccc|c} 6 & 3 & 2 & 18 \\ 0 & 33 & 0 & -132 \\ 0 & 0 & -2 & 6 \end{array} \right]$$

$$R_1 \leftarrow -11R_1 + R_2 \quad \left[\begin{array}{ccc|c} -66 & 0 & 0 & -330 \\ 0 & 33 & 0 & -132 \\ 0 & 0 & -2 & 6 \end{array} \right]$$

$$\begin{matrix} -1R_1 \\ -1R_3 \end{matrix} = \left[\begin{array}{ccc|c} 66 & 0 & 0 & 330 \\ 0 & 33 & 0 & -132 \\ 0 & 0 & 2 & -6 \end{array} \right]$$

$$\begin{matrix} R_1/66 \\ R_2/33 \\ R_3/2 \end{matrix} = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & -3 \end{array} \right]$$

$$\text{sol} \{ (5, -4, -3) \}$$